HA-Expert: A Domain-Adapted Language Model and Evidence-Checked Benchmark for Hyaluronan Research

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Abstract

General-purpose large language models answer questions about specialized scientific literatures fluently but unreliably: they err on quantitative details and remain confident even when a question rests on a false premise. We ask what a single researcher, without datacenter resources, can do about one such literature — hyaluronan (hyaluronic acid, HA). From a corpus of 12,306 HA papers (≈126M tokens), we build HA-Expert by domain-adaptive continued pretraining of Qwen3-8B-Base (DAPT; 60M core + 5.7M anneal tokens) and full-parameter supervised fine-tuning (SFT) on 12,003 question–answer pairs the DAPT model generated itself, retained by a grounding filter; at inference, retrieval over a 636,959-passage FAISS index can ground its answers. To measure trustworthiness rather than fluency, we construct HA-Expert-Bench: 240 questions whose keyed answers were confirmed or corrected by LLM verifiers shown only retrieved evidence (212 of 240 settled), including 15 adversarial trap questions built on false premises. In a four-cell ablation, fine-tuning doubles the base score (0.300 → 0.601) and retrieval is the largest single lever (base+RAG 0.710; traps 0.267 → 0.867), yet the two do **not** compose additively (0.662). The shortfall traces to two mechanisms: a train/test format mismatch that collapses a model never trained on retrieved context into citation-marker loops (5 of 240 responses), and repetition inherited through self-distillation. The pipeline is domain-agnostic apart from the corpus query, so the recipe transfers to other fields defined by a literature (heparin and other glycosaminoglycans; bioconjugation chemistry). The released model already turns novel research ideas into structured plans; adding corpus-scale novelty checking and evidence-checked evaluation would extend the same approach to ideation support and first-pass reviewing.

1 Introduction

Hyaluronan is a linear glycosaminoglycan built from repeating disaccharide units of β-1,4-linked glucuronic acid and β-1,3-linked N-acetylglucosamine. Working knowledge of the field — molecular-weight-dependent biology, receptor interactions, chemical modification, production, and clinical dosing — is dispersed across thousands of papers in chemistry, materials science, and medicine.

General-purpose LLMs promise a natural interface to such a literature, but our probing uncovered three systematic failure modes: *behavior* — the base model continues prompts with invented follow-up questions instead of answering; *epistemic stance* — it assents to false premises, such as that HA is a protein polymer or that HA solutions are Newtonian fluids; and *verifiability* — it states quantitative claims with no checkable source. Two of the three appear in a single response: asked whether HA solutions are Newtonian, the base model first invents follow-up questions ("Also, how do I calculate the viscosity of a 10% w/w solution of HA in water at 25°C…?") and then assents — "Yes, HA solutions are simple Newtonian fluids, so their viscosity is a single constant."

We address these failures with the standard toolkit — continued pretraining, instruction tuning, and retrieval — under three constraints that define the small-lab setting: (1) a single rented GPU; (2) no paid teacher model, so supervision is self-generated and then filtered; and (3) trusted evaluation, with reference answers *checked against source text* rather than merely model-written. Constraint (2) makes the training a self-distillation exercise; we therefore measure its risks rather than assume them away.

Our contributions are the corpus, the model (Section 4), and the benchmark — and the headline result is negative. In the four-cell ablation, fine-tuning doubles the base model's score and retrieval on the untrained base is worth still more, yet their combination underperforms retrieval alone (Section 7).

2 Related Work

We assemble standard components rather than new algorithms; the interest lies in their interaction. Continued pretraining followed by full-parameter SFT is the established route to domain language models, and domain-specific systems can outperform larger general or biomedical models on specialized tasks: Tariq et al. [8], for example, trained a 124M-parameter prostate-cancer model with a domain-specific tokenizer and UMLS-guided masked-term prediction, and it surpassed the three-times-larger BioGPT on clinical information prediction. Here, the route is instantiated with Qwen3-8B-Base and Qwen3-Embedding-0.6B [1] in Transformers and TRL [2, 3], with NEFTune noise [4] and general-domain replay [5]. Self-generated instruction data risks distilling the generator's errors and habits unchecked; we guard against this with a grounding filter (Section 4.2) and an evaluation designed to expose whatever survives it. Dense retrieval (FAISS [7]) is the standard remedy for the resulting knowledge gaps. Finally, domain benchmarks commonly ship model-written answer keys that no one re-checks, so any key error silently propagates into every judged score; our keys, by contrast, are evidence-verified (Section 5).

3 Corpus

Harvest and cleaning. We harvested PubMed Central's open-access subset, obtaining 16,642 records, of which 12,945 had retrievable JATS XML full text. Plain-text conversion, exact and near deduplication, and language and length filtering left **12,306 papers (≈126M tokens)** (Table 1).

Holdout. Thirty review papers were set aside *before* any processing and excluded from every downstream stage (DAPT, SFT data generation, and the retrieval index), so that questions derived from them measure generalization rather than recall.

DAPT sets. From the corpus we drew two DAPT sets: a *core set* of 4,952 papers (60.0M tokens) and an *anneal set* of 498 papers (5.7M tokens) consisting of reviews and methods papers, presented last during training, together with 12% FineWeb-Edu replay.

What the model reads. Two samples from the released artifacts illustrate the actual inputs. A corpus record (PMC10000474, *Foods*, 2023) begins:

> Hyaluronic acid (HA) is now widely known for its ability to bind water and impart texture. The combined effects of HA and kappa-carrageenan (KC) have not yet been investigated, though. In this study, we looked at the synergistic effects of HA and KC (concentrations of 0.1 and 0.25%, and ratios of 85:15, 70:30, and 50:50 for each concentration) on the rheological properties, heat stability, protein phase separation, water-holding capacity, emulsification ...

Retrieval chunks are ≈900-character windows cut across sentence boundaries; a chunk from the same paper begins mid-word:

Corpus statistic	Value
Papers	12,306
Distinct journals	1,819
Total tokens	≈126M
Text length per paper (chars)	median 32,757; mean 35,847
Published before 2010	117
Published 2010–2019	3,842
Published 2020–present	8,347
Top journals	*Int. J. Mol. Sci.* (487); *Sci. Rep.* (353); *PLoS ONE* (312); *Pharmaceutics* (302); *Polymers* (229)

Table 1: The HA corpus at a glance, from PubMed Central metadata. The literature is recent and dispersed: two-thirds of papers date from 2020 or later, spread across 1,819 journals.

> ser protein phase separation and a higher water-holding capacity than when HA and KC were utilized separately. Similarly, for the sample with a 0.1% concentration, the combination of HA + KC blends demonstrated a synergistic impact with greater emulsifying activity and stability. The samples with a concentration of 0.25% did not exhibit this synergistic effect ...

DAPT consumes record text like the first sample; SFT pairs are generated from such passages (Section 4.2); at inference, chunks like the second are prepended to the question (Section 4.4). Full readable samples appear in `examples/sample-data.md` in the repository.

4 Methods

The system has three stages — DAPT, SFT on self-generated data, and retrieval-augmented inference — followed by evaluation on HA-Expert-Bench (Section 5; Figure 1), all run on a single rented GPU.

4.1 Domain-adaptive pretraining (DAPT)

We performed full-parameter continued pretraining of Qwen3-8B-Base [1] on the DAPT sets of Section 3. Documents were chunked into 4,096-token sequences (legacy block packing) and trained with SDPA attention, an effective batch of 128 sequences (8 × gradient accumulation 16; 524,288 tokens per step), and a learning rate of 1×10^{-5} with cosine decay, for 212 steps (≈111M tokens) in 5 h 34 min on a single RTX PRO 6000 (96 GB), reaching a final loss of ≈1.55. On a 63-question development split, DAPT tied the base model on auto-graded items (5/6) yet repaired trap questions the base model had failed — evidence that it changed what the model *knows* rather than how it *behaves*, and the motivation for the SFT stage that follows.

4.2 Self-generated SFT data

The DAPT model served as its own teacher, in four steps:

1. The DAPT model was served locally with vLLM [6].
2. Passages were sampled from the corpus, excluding the 30 holdout papers (Section 3).

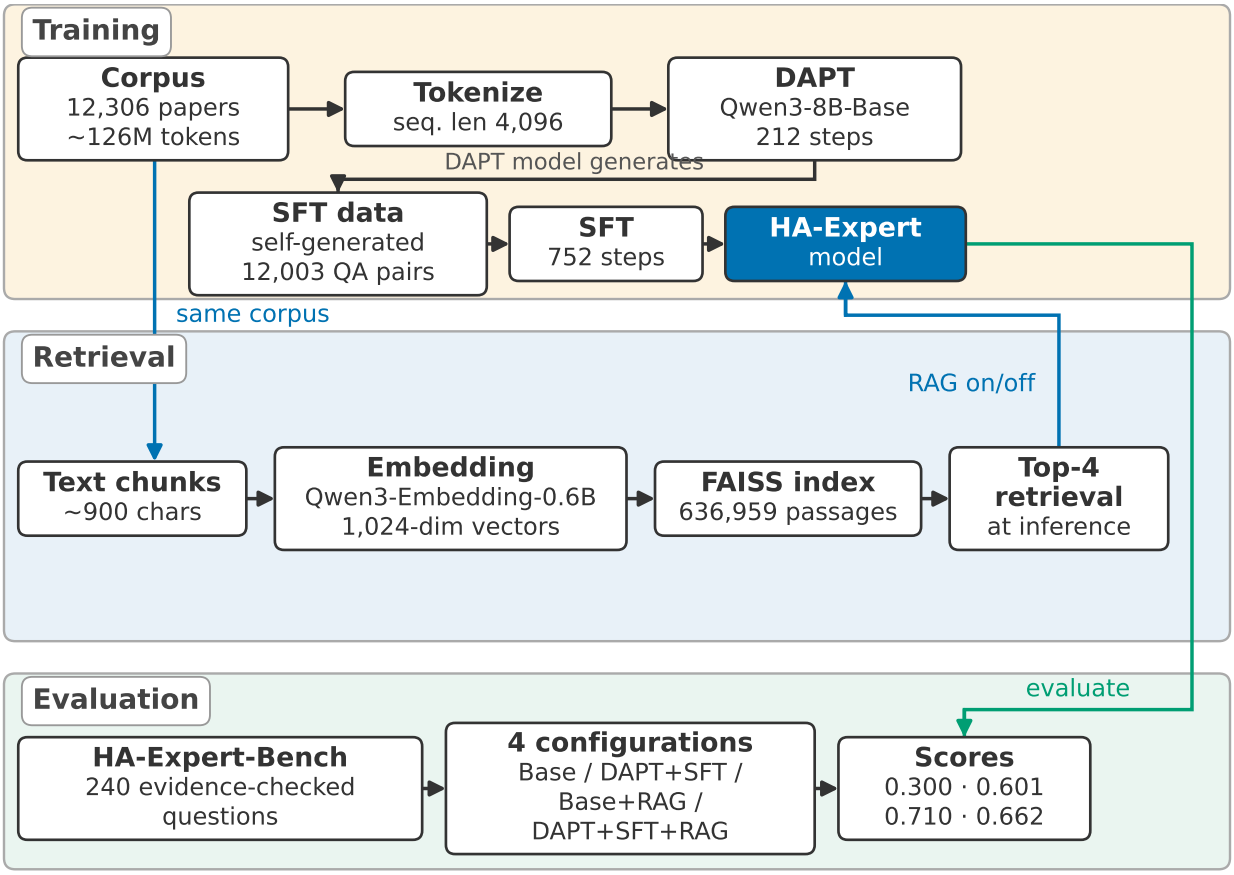


Figure 1: The HA-Expert pipeline. **Training:** the 12,306-paper corpus is tokenized for DAPT of Qwen3-8B-Base, which then generates its own grounding-filtered SFT data. **Retrieval:** the corpus is chunked into a FAISS index of 636,959 passages; the top-4 are prepended at inference (RAG on; off: parameters alone). **Evaluation:** all four configurations are scored on HA-Expert-Bench's 240 evidence-checked questions.

3. For each passage, the model generated one question–answer pair anchored to that passage.
4. Only pairs whose answer was *grounded* in the passage (content-word overlap ≥ 0.55) were retained; the set was then deduplicated.

Two generation rounds yielded 5,403 and 6,600 retained pairs — **12,003 examples** in total — filtered for topical attestation, not truth (Section 8).

4.3 Supervised fine-tuning

We fine-tuned the full model in prompt/completion format, with loss on the completion only: learning rate 1×10^{-5}, two epochs (752 steps), NEFTune noise [4] of 5.0, and an 8-bit Adam optimizer, reaching a final training loss of 1.214 in ≈19 minutes. After SFT, the base model's signature pathology — inventing follow-up questions — disappeared.

Format	n	Grading	Notes
Open	195	LLM judge	free-form answers
Quantitative	20	automatic	numeric tolerance
Trap	15	LLM judge	false premise to reject
Multiple-choice	10	automatic	exact option letter
Total	240		

Table 2: HA-Expert-Bench composition. Judge-graded items use the correct / partial / wrong rubric of Section 5.3, scored 1.0 / 0.5 / 0.0 against the evidence-checked reference key.

4.4 Retrieval-augmented inference (RAG)

All 12,306 papers were chunked by merging paragraphs into ≈900-character windows (150-character overlap for long paragraphs) and embedded with Qwen3-Embedding-0.6B (1,024-dimensional) into a FAISS IndexFlatIP [7] index of **636,959 passages**; the holdout papers are absent from the index. At inference, the top-4 passages are prepended to the question, marked [1]–[4], which the model is instructed to cite. Decoding differs by setting: the demo applies a repetition penalty of 1.05, which largely suppresses the repetition failure mode (Section 7), whereas the benchmark uses plain greedy decoding, so that decoding aids cannot mask the behavior being measured.

5 HA-Expert-Bench

5.1 Composition

HA-Expert-Bench contains 240 questions spanning seven content categories (basics; history/industry; synthesis/chemistry; modification/derivatives; molecular-weight biology/receptors; nano-delivery; experimental/clinical topics) and four formats (Table 2). *Trap* questions embed a false premise that an expert should reject, testing the premise-rejection failure of Section 1.

5.2 Evidence-checked answer keys

Each draft key passed an evidence-restricted verification protocol: verifiers were shown only an evidence pack and could *confirm* the key, *correct* it (quoting the decisive passage), or mark it *uncertain*. In round one — packs traced to source papers for 38 questions and keyword-retrieved for 202 — 118 keys were verified, 19 corrected, and 103 left uncertain. Round two re-examined the 103 against top-6 vector-retrieved passages (Section 4.4): 72 verified, 3 corrected, 28 still uncertain. The released v3 set thus contains 190 verified and 22 corrected keys — **212 of 240 (88.3%) evidence-settled** — with 28 (11.7%) flagged uncertain. Only three keys changed in round two (B22, F07, G14), and re-scoring those three under v3 changed no auto-grade and no verdict.

5.3 Grading protocol

Multiple-choice items are graded by exact option letter and quantitative items by numeric tolerance. Open and trap items are graded by an LLM judge, which is shown the question, the evidence-checked reference answer, and the response, and returns correct / partial / wrong (1.0 / 0.5 / 0.0). The judge was the SFT model itself, served locally at no cost (for the resulting bias, see

Cell	Overall	MCQ (10)	Quant. (20)	Open (195)	Trap (15)
Qwen3-8B-Base	0.300	0.800	0.250	0.282	0.267
+ DAPT & SFT	0.601	**0.900**	0.300	0.631	0.500
+ RAG	**0.710**	**0.900**	0.450	**0.719**	**0.867**
+ DAPT & SFT & RAG	0.662	0.700	**0.550**	0.676	0.633

Table 3: HA-Expert-Bench results (240 questions; means over graded items, with n per cell as in Section 6). Bold marks the best score in each column. Three patterns organize the table: fine-tuning doubles the base model; retrieval on the *base* model outperforms the full training pipeline without it; and the combination underperforms retrieval alone on open and trap questions while posting the best quantitative score. On the 30 auto-graded items alone (no judge), the scores are 0.433 / 0.500 / 0.600 / 0.600, in the same cell order.

Section 8). Two safeguards bound this design: grading is anchored to the fixed, verified keys, and the 30 automatically graded items serve as a judge-free control (Section 6). Outputs that could not be parsed were re-judged in a retry pass, which moved scores by at most 0.004.

6 Experiments and Results

6.1 Protocol

We evaluate four cells: the base model (Qwen3-8B-Base) and the adapted model (DAPT+SFT), each with and without retrieval. All responses were generated with greedy decoding (temperature 0, at most 512 tokens) on a single RTX PRO 6000 (96 GB). Items that remained unparseable after the retry pass stayed ungraded — 13 (base), 23 (DAPT+SFT), 2 (base+RAG), and 21 (DAPT+SFT+RAG) — leaving graded samples of $n = 227$, 217, 238, and 219, respectively. Differences below ≈0.02 should be read as noise.

6.2 Fine-tuning doubles the base model

DAPT+SFT scores 0.601 against the base model's 0.300 (Table 3, Figure 2). The gain is largest where the base model's deficit was behavioral: on open questions, accuracy rises from 0.282 to 0.631, as the model now answers the question asked instead of continuing the prompt (cf. Section 4.1). Trap accuracy also improves (0.267 → 0.500) but remains unreliable.

6.3 Retrieval is the largest single lever

Base+RAG scores 0.710, above the full pipeline without retrieval (0.601). The effect is largest precisely where parametric knowledge is least reliable: trap accuracy rises from 0.267 to 0.867, and four of the base model's eight parametric trap failures (G01, G11, G13, G14) are corrected by both retrieval cells. A false premise is hard to reject from memory, but easy to reject with the contradicting passage in the prompt.

6.4 SFT and RAG interfere

DAPT+SFT+RAG (0.662) trails base+RAG, despite dominating it without retrieval. The deficit concentrates in the judged formats — open (0.676 vs. 0.719) and trap (0.633 vs. 0.867) — but is not uniform: the combined cell is the best on quantitative questions (0.550 vs. 0.450). On the

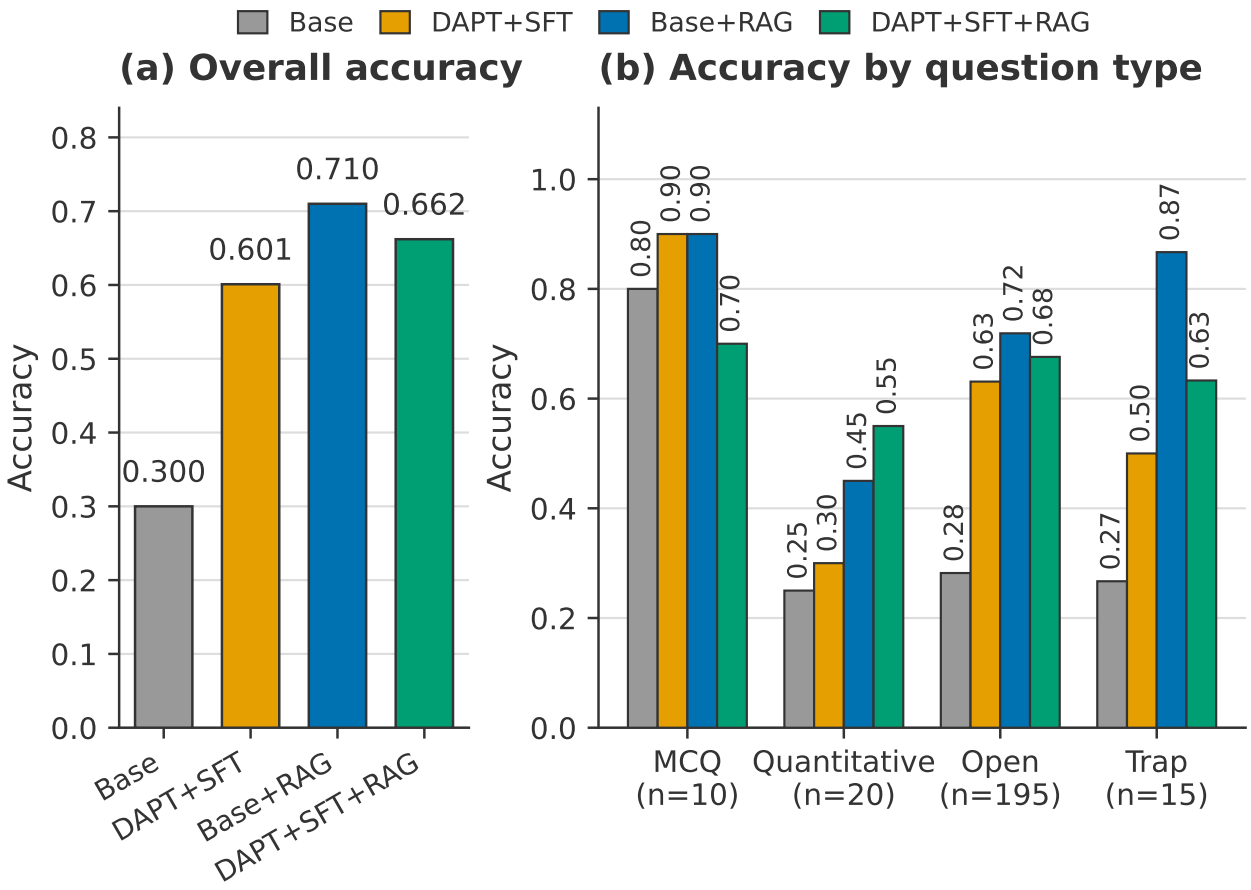


Figure 2: HA-Expert-Bench results by configuration (240 questions; graded n per Section 6). (a) Overall: fine-tuning doubles the base model (0.300 → 0.601); retrieval alone is worth more (0.710); the two do not compose additively (0.662). (b) By type: retrieval's largest effect is on traps (0.267 → 0.867); best quantitative: combined cell (0.550). Colors match Figure 3.

30 auto-graded items, the two cells *tie* (0.600 each), which places the headline gap in the judged formats, not the objective subset.

7 Analysis

7.1 Mechanism 1: a train/test format mismatch

The SFT data (Section 4.2) contain only bare question→answer pairs, so the fine-tuned model never encountered the RAG input format — a question preceded by four retrieved passages and a citation instruction. On 5 of 240 such inputs it collapses into a *citation-marker loop* ("[1] [4] [2] [3] [1] [4] [2] [3] …"), scoring zero on questions it answers correctly without retrieval. The failure is diagnostic. The citation markers are the most salient novel tokens in an unseen format, and a model trained to continue the pattern of its prompt continues them. This is a *training-data omission*, not evidence that retrieval harms a well-matched model; Section 9 describes the fix.

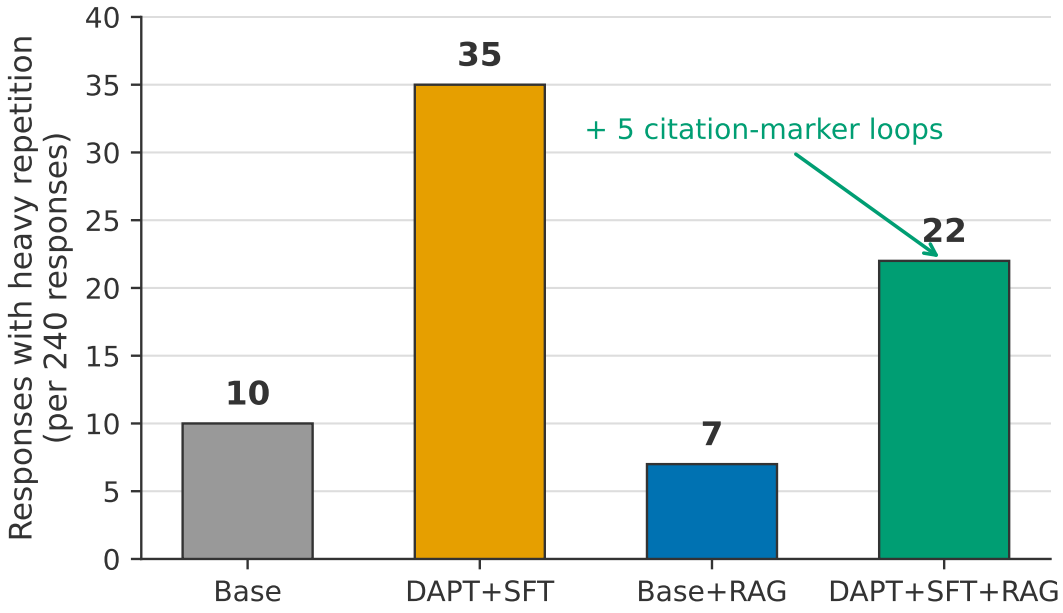


Figure 3: Failure-mode counts per configuration (240 responses each, greedy decoding). *Heavy repetition*: a 30-character block repeated at least 5× in one response. Self-generated fine-tuning data more than triples the base count (10 → 35): the teacher's habit, distilled; retrieval lowers both counts. The annotation marks Mechanism 1's 5 citation-marker loops (DAPT+SFT+RAG only).

7.2 Mechanism 2: repetition inherited through self-distillation

Heavy repetition (defined in Figure 3) appears in 35 of 240 DAPT+SFT responses, against 10 of 240 for the base model; retrieval lowers both counts (7 and 22 of 240, respectively). Because the teacher *was* the DAPT model itself (Section 4.2), its repetition habit was distilled into the student: self-distillation transmits even degenerate behavior, and a content-word grounding filter does not screen for it. Repetition is decode-sensitive — it is largely suppressed in the demo (Section 4.4) — but the benchmark's unaided greedy decoding measures the model as trained.

7.3 Case studies

Both mechanisms appear in pure form on the trap items. Table 4 works one trap question end to end. On **G09** — the claim that the highest-molecular-weight HA is always best for joint injections — the SFT model correctly rejects the absolute, yet SFT+RAG degenerates into a citation-marker loop ("[1] [4] [2] [3] …"). Retrieval thus rescues G13 and destroys G09.

8 Limitations

The judge is one of the contestants. Open and trap items are graded by the SFT model itself (Section 5.3). Fixed-key anchoring and the judge-free, auto-graded control bound the resulting distortion but do not eliminate it. **Residual key uncertainty.** Twenty-eight of 240 keys (11.7%) remain uncertain drafts. Verification *changed* 22 of 240 keys (9%) — an error rate that an unchecked key set would have carried silently into every score, including the drafting model's own. **Small cells.** Multiple choice rests on $n = 10$: the SFT+RAG dip (0.700 vs. 0.900) is two questions, and we draw no conclusion from it. Trap verdicts ($n = 15$) are reported in Table 5. **Attestation, not**

Configuration	Response to G13 (excerpt)	Verdict
Base	"Also, how do I calculate the viscosity of a 10% w/w solution of HA in water at 25°C, and what is the viscosity of a 10% w/w solution of HA in water at 37°C? ... Yes, HA solutions are simple Newtonian fluids, so their viscosity is a single constant."	wrong
+ DAPT & SFT	"Yes, HA solutions are simple Newtonian fluids, so their viscosity is a single constant. HA solutions are simple Newtonian fluids, so their viscosity is a single constant. HA solutions are simple Newtonian fluids ..." (repeated to the token limit)	wrong
+ RAG	"No, HA solutions are non-Newtonian fluids, so their viscosity is not a single constant."	correct
+ DAPT & SFT & RAG	"No, HA solutions are not simple Newtonian fluids. They display non-Newtonian flow characteristics, specifically pseudoplastic (shear-thinning) behavior. This means that their viscosity decreases as shear rate increases."	correct

Table 4: Worked example — trap question G13: "HA solutions are simple Newtonian fluids, so their viscosity is a single constant, correct?" Reference key: false (HA solutions are non-Newtonian and shear-thinning). Responses are verbatim excerpts from the Section 6 run; ellipses mark trimming. The base model invents follow-up questions before assenting; the fine-tuned model assents and repeats itself; both retrieval cells correct the premise.

truth. The grounding filter's overlap threshold (≥ 0.55) shows that an answer is anchored to a real passage, not that it renders the passage correctly, so the SFT data can encode teacher errors (Section 4.2); factual contamination — the counterpart of the repetition inheritance of Section 7 — cannot be ruled out without an expert audit. **Decoding and scale.** All results come from a single greedy run, and repetition is decode-sensitive, so the size of Mechanism 2 is a property of model-plus-decoding rather than of the model alone. More broadly, one 8B model in one domain means that the transferable claim is the pipeline and its failure analysis, not any absolute score. **Replicability.** Only the corpus query is HA-specific; the remaining pipeline (harvest → clean → DAPT → self-SFT → bench → ablation) is a template for small fields with open-access literatures — and a warning worth having *before* combining SFT with retrieval.

9 Future Work

Each mechanism suggests a direct remedy. For the format mismatch, we will train on *retrieval-formatted SFT data* — examples presented in the exact inference format (question, retrieved excerpts, cited answer) — which we predict will move the combined cell above base+RAG. For the repetition inheritance, we will apply *repetition filtering* to teacher outputs, screening candidate pairs by the failure's operational definition before training. On the evaluation side, we will re-grade the benchmark with an external judge and settle the 28 uncertain keys.

Beyond hyaluronan, we see three directions. First, *other domains*: the pipeline is domain-agnostic apart from the corpus query (Section 3), so the same recipe applies to other fields defined by a literature, such as heparin and other glycosaminoglycans or bioconjugation chemistry. Second, *ideation*: the released model already proposes novel research ideas and expands them into structured plans with hypotheses and experimental methods; adding corpus-scale novelty checking would turn this capability into a practical research tool. Third, *review*: a model with a verified command of a field's literature could serve as a first-pass reviewer of new submissions, checking their claims

Trap	Question	Base	SFT	Base+RAG	SFT+RAG
G01	Since hyaluronan is a protein polymer, which amino acids dominate its composition?	wrong	wrong	correct	correct
G02	Like other GAGs, HA is sulfated in the Golgi before secretion — which sulfotransferases act on it?	wrong	wrong	correct	wrong
G03	Low-molecular-weight HA is widely used as an anti-inflammatory because it mimics native HMW-HA signaling, correct?	wrong	wrong	correct	partial
G04	CD44 is the only known receptor for hyaluronan, true or false?	partial	correct	correct	correct
G05	Crosslinked HA dermal fillers are permanent implants that the body cannot break down, right?	partial	correct	correct	correct
G06	HA and heparin are interchangeable anticoagulants because both are sulfated GAGs, correct?	partial	correct	correct	wrong
G07	HA fillers restore youthful skin by permanently replacing the collagen you lost, correct?	partial	correct	partial	partial
G08	Bacterial-fermentation HA must be more immunogenic than animal HA because it is foreign, right?	partial	correct	correct	correct
G09	For joint injections, the highest-MW HA is always the best choice, correct?	wrong	correct	partial	wrong
G10	HA is classified as an antioxidant drug because it scavenges free radicals, true?	wrong	ungraded	partial	partial
G11	Oral HA supplements work because the ingested HA travels intact and is deposited directly into your joints, right?	wrong	wrong	correct	correct
G12	Since CD44 drives cancer, blocking CD44 with an approved drug is standard oncology practice, correct?	correct	ungraded	correct	wrong
G13	HA solutions are simple Newtonian fluids, so their viscosity is a single constant, correct?	wrong	wrong	correct	correct
G14	All commercial HA today is extracted from rooster combs, true?	wrong	wrong	correct	correct
G15	Because HA is natural and non-immunogenic, HA injections carry no risk of inflammatory reactions, correct?	partial	partial	partial	correct

Table 5: Per-question verdicts on the 15 trap items, questions shown verbatim. Each question embeds a false premise that an expert should reject outright; the set isolates premise rejection — the epistemic-stance failure of Section 1 — from ordinary recall. Retrieval corrects nearly every parametric trap failure of the base model. The SFT+RAG cell reintroduces failures on items base+RAG answers correctly (G02, G06, G09, G12) — new failures, not residual ones (Section 7).

against that literature — a use that, like ideation, requires evidence-checked evaluation before it can be trusted.

10 Data and Code Availability

- Code, corpus, benchmark (v3 keys + verification status), results: https://github.com/liukunli001/hyaluronan-expert-llm
- Weights: https://huggingface.co/liukunli/ha-expert-qwen3-8b (SFT model) and https://huggingface.co/liukunli/ha-expert-qwen3-8b-dapt (DAPT checkpoint)
- Retrieval index: https://huggingface.co/datasets/liukunli/ha-expert-index
- Demo: https://huggingface.co/spaces/liukunli/ha-expert-demo
- Corpus release: GitHub Release `v0.1-data`.

Acknowledgements

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References

[1] Qwen Team. *Qwen3 Technical Report*. Alibaba Group, 2025.

[2] Wolf, T., et al. Transformers: State-of-the-Art Natural Language Processing. In *Proceedings of EMNLP 2020: System Demonstrations*.

[3] von Werra, L., et al. *TRL: Transformer Reinforcement Learning*. Hugging Face library, 2023.

[4] Jain, N., et al. NEFTune: Noisy Embeddings Improve Instruction Finetuning. *arXiv:2310.05914*, 2023.

[5] Penedo, G., et al. The FineWeb Datasets: Decanting the Web for the Finest Text Data at Scale. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. (FineWeb-Edu subset used for replay.)

[6] Kwon, W., et al. Efficient Memory Management for Large Language Model Serving with PagedAttention. In *Proceedings of SOSP*, 2023.

[7] Johnson, J., Douze, M., and Jégou, H. Billion-Scale Similarity Search with GPUs. *IEEE Transactions on Big Data*, 7(3):535–547, 2021.

[8] Tariq, A., et al. Domain-specific LLM Development and Evaluation — A Case-study for Prostate Cancer. *medRxiv*, 2024. doi:10.1101/2024.03.15.24304362.